URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.551(108).pdf
RESOLUTION MSC.551(108) (adopted on 23 May 2024)
AMENDMENTS TO THE INTERNATIONAL CODE OF SAFETY
FOR SHIPS USING GASES OR OTHER LOW-FLASHPOINT FUELS (IGF CODE)
RESOLUTION MSC.551(108)
(adopted on 23 May 2024)
AMENDMENTS TO THE INTERNATIONAL CODE OF SAFETY
FOR SHIPS USING GASES OR OTHER LOW-FLASHPOINT FUELS (IGF CODE)
THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,
NOTING resolution MSC.391(95), by which it adopted the International Code of Safety for Ships using Gases or other Low -flashpoint Fuels (IGF Code), which has become mandatory under chapters II-1 and II-2 of the International Convention for the Safety of Life at Sea, 1974 ("the Convention"),
NOTING ALSO article VIII(b) and regulation II -1/2.28 of the Convention concerning the procedure for amending the IGF Code,
HAVING CONSIDERED, at its 108th session, amendments to the IGF Code proposed and
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procedure for amending the IGF Code,
HAVING CONSIDERED, at its 108th session, amendments to the IGF Code proposed and circulated in accordance with article VIII(b)(i) of the Convention:
1 ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to the IGF Code, the text of which is set out in the annex to the present resolution;
2 DETERMINES, in accordance with article VIII(b)(vi)(2)(bb) of the Convention, that the said amendments shall be deemed to have been accepted on 1 July 2025, unless, prior to that date, more than one third of the Contracting Governments to the Convention or Contracting Governments the combined merchant fleets of which constitute not less than 50% of the gross tonnage of the world's merchant fleet have notified the Secretary -General of their objections to the amendments;
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4 REQUESTS the Secretary -General, for the purposes of article VIII(b)(v) of the Convention, to transmit certified copies of the present resolution and the text of the amendments contained in the annex to all Contracting Governments to the Convention;
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ANNEX
AMENDMENTS TO THE INTERNATIONAL CODE OF SAFETY
FOR SHIPS USING GASES OR OTHER LOW-FLASHPOINT FUELS (IGF CODE)
Part A

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ANNEX

AMENDMENTS TO THE INTERNATIONAL CODE OF SAFETY
FOR SHIPS USING GASES OR OTHER LOW-FLASHPOINT FUELS (IGF CODE)

Part A

2 General

2.2 Definitions

1 The following new paragraph 2.2.43 is added after existing paragraph 2.2.42:

"2.2.43 Ship constructed on or after 1 January 2026 means:

.1 for which the building contract is placed on or after 1 January 2026;
or

.2 in the absence of a building contract, the keels of which are laid or which are at a similar stage of construction on or after 1 July 2026;
or

.3 the delivery of which is on or after 1 January 2030."

4 General requirements

4.2 Risk assessment

2 Paragraph 4.2.2 is replaced by the following:

"4.2.2 For ships to which part A-1 applies, the risk assessment required by 4.2.1 need only be conducted where explicitly required by paragraphs 5.10.5, 5.12.3, 6.4.1.1, 6.4.15.4.7.2, 8.3.1.1, 8.4.2, 13.4.1, 13.7 and 15.8.1.10 as well as by paragraphs 4.4 and 6.8 of the annex."

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paragraphs 4.4 and 6.8 of the annex."
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3 Paragraph 5.3.3.3 is replaced by the following:
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6 Fuel containment system
6.4 Regulations for liquefied gas fuel containment
6.4.15 Tank types
6.4.15.3 Type C independent tanks
6.4.15.3.1 Design basis
6 Paragraph 6.4.15.3.1.2 is replaced by the following:
"6.4.15.3.1.2 The design vapour pressure shall not be less than:

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6.4.15.3.1 Design basis
6 Paragraph 6.4.15.3.1.2 is replaced by the following:
"6.4.15.3.1.2 The design vapour pressure shall not be less than:
0 = 0.2 + () 1.5 (MPa)
where:
 = 0.00185 (/) 2
with:
 = design primary membrane stress;
 = allowable dynamic membrane stress (double amplitude at probability level Q = 10-8) and equal to:
- 55 N/mm2 for ferritic-perlitic, martensitic and austenitic steel;
- 25 N/mm2 for aluminium alloy (5083-O);
 = a characteristic tank dimension to be taken as the greatest of the following:
h, 0.75b or 0.45,
with:
h = height of tank (dimension in ship's vertical direction) (m);
b = width of tank (dimension in ship's transverse direction) (m);
 = length of tank (dimension in ship's longitudinal direction) (m);
 = the relative density of the fuel (= 1 for fresh water) at the design temperature."
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design temperature."
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6.7 Regulations for pressure relief system
6.7.3 Sizing of pressure relieving system
6.7.3.1 Sizing of pressure relief valves
7 The chapeau of paragraph 6.7.3.1.1 is replaced by the following:
"6.7.3.1.1 For ships constructed on or after 1 January 2026, the pressure relief system for each liquefied gas fuel tank shall be designed so that, regardless of the state of any one PRV, the capacity of the residual PRVs meets the combined relieving capacity requirements of the system. The combined relieving capacity shall be the greater of the following, with no more than 20% rise in liquefied gas fuel tank pressure above the MARVS. The tank shall not be loaded until the full relieving capacity is restored:"
8 Paragraph 6.7.3.1.1.2 is replaced by the following:

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restored:"
8 Paragraph 6.7.3.1.1.2 is replaced by the following:
"6.7.3.1.1.2 vapours generated under fire exposure computed using the
following formula:
 = 0.82 (m3/s)
where:
 = minimum required rate of discharge of air at standard conditions
of 273.15 Kelvin (K) and 0.1013 MPa.
 = fire exposure factor for different liquefied gas fuel tank types:
 = 1.0 for tanks without insulation located on deck;
"
6.9 Regulations for the maintaining of fuel storage condition
6.9.1 Control of tank pressure and temperature
9 The chapeau of paragraph 6.9.1.1 is replaced by the following:
"6.9.1.1 For ships constructed on or after 1 January 2026, w ith the exception of
liquefied gas fuel tanks designed to withstand the full gauge vapour pressure of the
fuel under conditions of the upper ambient design temperature, liquefied gas fuel
tanks' pressure and temperature shall be maintained at all times wit hin their design
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tanks' pressure and temperature shall be maintained at all times wit hin their design
range by means acceptable to the Administration, e.g. by one or more of the following
methods:"
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- 5 -
7 Material and general pipe design
7.3 Regulations for general pipe design
7.3.2 Wall thickness
10 Paragraph 7.3.2.1 is replaced by the following:
"7.3.2.1 For ships constructed on or after 1 January 2026, the minimum wall thickness
shall be calculated as follows:
t = (t0 + b + c) / (1 ||/100) (mm)
where:
t0 = theoretical thickness
t0 = PD / (2.0Ke + P) (mm)
with:
P = design pressure (MPa) referred to in 7.3.3;
D = outside diameter (mm);
K = allowable stress (N/mm²) referred to in 7.3.4; and
 e = efficiency factor equal to 1.0 for seamless pipes and for
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K = allowable stress (N/mm²) referred to in 7.3.4; and

e = efficiency factor equal to 1.0 for seamless pipes and for longitudinally or spirally welded pipes, delivered by approved manufacturers of welded pipes, that are considered equivalent to seamless pipes when non -destructive testing on welds is carried out in accordance with recognized standards. In other cases an efficiency factor of less than 1.0, in accordance with recognized standards, may be required depending on the manufacturing process;

b = allowance for bending (mm). The value of b shall be chosen so that the calculated stress in the bend, due to internal pressure only, does not exceed the allowable stress. Where such justification is not given, b shall be:

b = Dt0 / 2.5r (mm)

with:

r = mean radius of the bend (mm);

c = corrosion allowance (mm). If corrosion or erosion is expected the wall thickness of the piping shall be increased over that required by other

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thickness of the piping shall be increased over that required by other
design regulations. This allowance shall be consistent with the
expected life of the piping; and
= negative manufacturing tolerance for thickness (%), i.e. where is the
manufacturing tolerance of -5%, || is equal to 5 and shall be entered
into the formula as 1- (5/100)."
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- 6 -
8 Bunkering
8.4 Regulations for manifold
11 Paragraph 8.4.1 is replaced by the following:
"8.4.1 The bunkering manifold shall be designed to withstand the external loads
during bunkering. The connections at the bunkering station shall be arranged in order
to achieve a dry-disconnect operation in one of the followings ways:
.1 a dry-disconnect / connect coupling in accordance with a standard
at least equivalent to those acceptable to the Organization; or
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.1 a dry-disconnect / connect coupling in accordance with a standard at least equivalent to those acceptable to the Organization; or

.2 a manual connect coupler or hydraulic connect coupler, used to connect the bunker system to the receiving vessel bunkering manifold presentation flange; or

.3 a bolted flange to flange assembly.

12 The following new paragraphs are added after existing paragraph 8.4.1:

8.4.2 When intended to use either of the connections specified in paragraphs 8.4.1.2 and 8.4.1.3, these shall be combined with operating procedures that ensure a dry-disconnect is achieved. The arrangement shall be subject to special consideration informed by a bunkering arrangement risk assessment conducted at the design stage and considering dynamic loads at the bunkering manifold connection to a recognized standard acceptable to the Administration, the safe operation of the ship and other hazards that may be r elevant to the ship during bunkering operation.

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ship and other hazards that may be r elevant to the ship during bunkering operation.
The fuel handling manual required by 18.2.3 shall include documentation that the bunkering arrangement risk assessment was conducted, and that special consideration was granted under this requirement.
8.4.3 An emergency release coupler (ERC) / Emergency Release System (ERS) or equivalent means shall be provided, unless installed on the bunkering supply side of the bunkering line, and the said means shall be in accordance with a standard equivalent to those acceptable to the Organization; it shal l enable a quick physical disconnection dry break-away of the bunker system in an emergency event."
9 Fuel supply to consumers
9.3 Regulations on redundancy of fuel supply
13 Paragraph 9.3.1 is replaced by the following:
"9.3.1 For ships constructed on or after 1 January 2026, for single fuel installations the fuel supply system shall be arranged with redundancy and segregation, so that a
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the fuel supply system shall be arranged with redundancy and segregation, so that a leakage in one system, or failure of one of the fuel supply essential auxiliaries, does not lead to an unacceptable loss of power. In the event of a leakage or failure, and in accordance with SOLAS regulation II -1/26.3, the Administration, having regard to overall safety considerations, may accept a partial reduction in propulsion capabili ty from normal operation."

9.4 Regulations on safety functions of gas supply system

14 Paragraph 9.4.7 is replaced by the following:

"9.4.7 For ships constructed on or after 1 January 2026, in cases where the master gas fuel valve is automatically shut down when the safety system as required in 15.2.2 is activated, the complete gas supply pipe between this master gas fuel valve and the

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is activated, the complete gas supply pipe between this master gas fuel valve and the double block and bleed valves and between the double block and bleed valves and the consumer shall be automatically vented."

15 Paragraph 9.4.8 is replaced by the following:

"9.4.8 For ships constructed on or after 1 January 2026, there shall be one manually operated shutdown valve in the gas supply line to each gas consumer upstream of the double block and bleed valves to assure safe isolation during maintenance on the gas consumer."

9.6 Regulations for fuel supply to consumers in gas-safe machinery spaces

16 Paragraph 9.6.1.1 is replaced by the following:

"9.6.1 Gas fuel piping in gas-safe machinery spaces shall be completely enclosed by a double pipe or duct fulfilling one of the following conditions:

.1 the gas fuel piping shall be a double wall piping system with the gas fuel contained in the inner pipe. The space between the concentric

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.1 the gas fuel piping shall be a double wall piping system with the gas fuel contained in the inner pipe. The space between the concentric pipes shall be pressurized with inert gas at a pressure greater than the gas fuel pressure. Suitable alarms shall be provided to indicate a loss of inert gas pressure between the pipes; or"

9.8 Regulations for the design of ventilated duct, outer pipe against inner pipe gas leakage

17 Paragraph 9.8.1 is replaced by the following:

"9.8.1 For ships constructed on or after 1 January 2026, the design pressure of the outer pipe or duct of fuel systems shall not be less than the maximum working pressure of the inner pipe. Alternatively, the design pressure of the outer pipe or duct may be calculated in accordance with 9.8.2."

18 The chapeau of paragraph 9.8.2 is replaced by the following:

"9.8.2 For ships constructed on or after 1 January 2026, alternatively to 9.8.1, the

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"9.8.2 For ships constructed on or after 1 January 2026, alternatively to 9.8.1, the design pressure of the outer pipe or duct shall be taken as the higher of the following:"

19 Paragraph 9.8.4 is replaced by the following:

"9.8.4 For ships constructed on or after 1 January 2026, the duct shall be pressure-tested to show that it can withstand the expected maximum pressure at fuel pipe rupture."

11 Fire safety

11.3 Regulations for fire protection

20 Paragraph 11.3.1 is replaced by the following:

"11.3.1 For ships constructed on or after 1 January 2026, fuel preparation rooms shall, for the purpose of the application of SOLAS regulation II-2/9, be regarded as a machinery space of category A."

11.6 Regulations for dry chemical powder fire-extinguishing system

21 Paragraph 11.6.2 is replaced by the following:

11.6 Regulations for dry chemical powder fire-extinguishing system
21 Paragraph 11.6.2 is replaced by the following:
"11.6.2 In addition to any other portable fire extinguishers that may be required elsewhere in IMO instruments, one portable dry powder extinguisher of at least 5 kg capacity shall be located near the bunkering station and in the fuel preparation room. For ships constructed before 1 January 2026, the portable dry powder extinguisher shall be provided in the fuel preparation room not later than the first survey on or after 1 January 2026."
12 Explosion prevention
12.5 Hazardous area zones
22 Paragraph 12.5.1 is replaced by the following:
"12.5.1 Hazardous area zone 0
For ships constructed on or after 1 January 2026, this zone includes, but is not limited to, the interiors of fuel tanks, any pipework for pressure relief or other venting systems for fuel tanks, pipes and equipment containing fuel, and interbarrier spaces as defined
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for fuel tanks, pipes and equipment containing fuel, and interbarrier spaces as defined by paragraph 2.2.15.2."

12.5.2 Hazardous area zone 1

23 Paragraph 12.5.2.1 is replaced by the following:

".1 for ships constructed on or after 1 January 2026, tank connection spaces and fuel storage hold spaces;"

15 Control, monitoring and safety systems

15.4 Regulations for bunkering and liquefied gas fuel tank monitoring

15.4.1 Level indicators for liquefied gas fuel tanks

24 Paragraph 15.4.1.3 is replaced by the following:

".3 For ships constructed on or after 1 January 2026, liquefied gas fuel tank liquid level gauges may be of the following types:

.1 indirect devices which determine the amount of fuel by means such as weighing or in-line flow metering;

.2 closed devices which do not penetrate the liquefied gas fuel tank, such as devices using radioisotopes or ultrasonic devices; or

.3 closed devices which penetrate the liquefied gas fuel tank but which

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such as devices using radioisotopes or ultrasonic devices; or
.3 closed devices which penetrate the liquefied gas fuel tank but which form part of a closed system and keep the gas fuel from being released. Such devices shall be considered as tank connections. If the closed gauging device is not mounted directly onto th e tank, it shall be provided with a shut off valve located as close as possible to the tank."

Part B-1

16 Manufacture, workmanship and testing

16.3 Welding of metallic materials and non -destructive testing for the fuel containment system

16.3.5 Production weld tests

25 Paragraph 16.3.5.1 is replaced by the following:

"16.3.5.1 For all fuel tanks and process pressure vessels except membrane tanks, production weld tests shall generally be performed for approximately

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tanks, production weld tests shall generally be performed for approximately each 50 m of butt -weld joints and shall be representative of each welding position. For secondary barriers, the same type production tests as required for primary barriers shall be performed, except that the number of tests may be reduced subject to agreement with the Administration. Tests, other than those specified in 16.3.5.2 to 16.3.5.5, may be required for fuel tanks or secondary barriers."

Part C-1

18 Operation

18.4 Regulations for bunkering operations

18.4.1 Responsibilities

26 Paragraph 18.4.1.1.1 is replaced by the following:

"18.4.1.1 Before any bunkering operation commences, the master of the receiving ship or their representative and the representative of the bunkering source (Persons In Charge, PIC) shall:

.1 agree in writing the transfer procedure, including cooling down and if necessary, gassing up; the maximum transfer rate at all stages ;

.1 agree in writing the transfer procedure, including cooling down and
if necessary, gassing up; the maximum transfer rate at all stages ;
minimum and maximum limiting transfer pressure and temperature;
bunkering line PRVs settings; and volume to be transferred;"
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6 Paragraph 6.4.15.3.1.2 is replaced by the following:
"6.4.15.3.1.2 The design vapour pressure shall not be less than:

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6.4.15.3.1 Design basis

6 Paragraph 6.4.15.3.1.2 is replaced by the following:

"6.4.15.3.1.2 The design vapour pressure shall not be less than:

0 = 0.2 + () 1.5 (MPa)

where:

 = 0.00185 (/) 2

with:

 = design primary membrane stress;

 = allowable dynamic membrane stress (double amplitude at probability level Q = 10-8) and equal to:

- 55 N/mm2 for ferritic-perlitic, martensitic and austenitic steel;
- 25 N/mm2 for aluminium alloy (5083-O);

 = a characteristic tank dimension to be taken as the greatest of the following:

h, 0.75b or 0.45,

with:

h = height of tank (dimension in ship's vertical direction) (m);

b = width of tank (dimension in ship's transverse direction) (m);

 = length of tank (dimension in ship's longitudinal direction) (m);

 = the relative density of the fuel (= 1 for fresh water) at the design temperature."

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design temperature."
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6.7 Regulations for pressure relief system
6.7.3 Sizing of pressure relieving system
6.7.3.1 Sizing of pressure relief valves
7 The chapeau of paragraph 6.7.3.1.1 is replaced by the following:
"6.7.3.1.1 For ships constructed on or after 1 January 2026, the pressure relief system for each liquefied gas fuel tank shall be designed so that, regardless of the state of any one PRV, the capacity of the residual PRVs meets the combined relieving capacity requirements of the system. The combined relieving capacity shall be the greater of the following, with no more than 20% rise in liquefied gas fuel tank pressure above the MARVS. The tank shall not be loaded until the full relieving capacity is restored:"
8 Paragraph 6.7.3.1.1.2 is replaced by the following:
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restored:"
8 Paragraph 6.7.3.1.1.2 is replaced by the following:
"6.7.3.1.1.2 vapours generated under fire exposure computed using the
following formula:
 = 0.82 (m3/s)
where:
 = minimum required rate of discharge of air at standard conditions
of 273.15 Kelvin (K) and 0.1013 MPa.
 = fire exposure factor for different liquefied gas fuel tank types:
 = 1.0 for tanks without insulation located on deck;
"
6.9 Regulations for the maintaining of fuel storage condition
6.9.1 Control of tank pressure and temperature
9 The chapeau of paragraph 6.9.1.1 is replaced by the following:
"6.9.1.1 For ships constructed on or after 1 January 2026, w ith the exception of
liquefied gas fuel tanks designed to withstand the full gauge vapour pressure of the
fuel under conditions of the upper ambient design temperature, liquefied gas fuel
tanks' pressure and temperature shall be maintained at all times wit hin their design
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tanks' pressure and temperature shall be maintained at all times wit hin their design
range by means acceptable to the Administration, e.g. by one or more of the following
methods:"
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7 Material and general pipe design
7.3 Regulations for general pipe design
7.3.2 Wall thickness
10 Paragraph 7.3.2.1 is replaced by the following:
"7.3.2.1 For ships constructed on or after 1 January 2026, the minimum wall thickness
shall be calculated as follows:
t = (t0 + b + c) / (1 ||/100) (mm)
where:
t0 = theoretical thickness
t0 = PD / (2.0Ke + P) (mm)
with:
P = design pressure (MPa) referred to in 7.3.3;
D = outside diameter (mm);
K = allowable stress (N/mm²) referred to in 7.3.4; and
 e = efficiency factor equal to 1.0 for seamless pipes and for
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K = allowable stress (N/mm²) referred to in 7.3.4; and

e = efficiency factor equal to 1.0 for seamless pipes and for longitudinally or spirally welded pipes, delivered by approved manufacturers of welded pipes, that are considered equivalent to seamless pipes when non -destructive testing on welds is carried out in accordance with recognized standards. In other cases an efficiency factor of less than 1.0, in accordance with recognized standards, may be required depending on the manufacturing process;

b = allowance for bending (mm). The value of b shall be chosen so that the calculated stress in the bend, due to internal pressure only, does not exceed the allowable stress. Where such justification is not given, b shall be:

b = Dt0 / 2.5r (mm)

with:

r = mean radius of the bend (mm);

c = corrosion allowance (mm). If corrosion or erosion is expected the wall thickness of the piping shall be increased over that required by other

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thickness of the piping shall be increased over that required by other
design regulations. This allowance shall be consistent with the
expected life of the piping; and
= negative manufacturing tolerance for thickness (%), i.e. where is the
manufacturing tolerance of -5%, || is equal to 5 and shall be entered
into the formula as 1- (5/100)."
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8 Bunkering
8.4 Regulations for manifold
11 Paragraph 8.4.1 is replaced by the following:
"8.4.1 The bunkering manifold shall be designed to withstand the external loads
during bunkering. The connections at the bunkering station shall be arranged in order
to achieve a dry-disconnect operation in one of the followings ways:
.1 a dry-disconnect / connect coupling in accordance with a standard
at least equivalent to those acceptable to the Organization; or
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.1 a dry-disconnect / connect coupling in accordance with a standard at least equivalent to those acceptable to the Organization; or

.2 a manual connect coupler or hydraulic connect coupler, used to connect the bunker system to the receiving vessel bunkering manifold presentation flange; or

.3 a bolted flange to flange assembly.

12 The following new paragraphs are added after existing paragraph 8.4.1:

8.4.2 When intended to use either of the connections specified in paragraphs 8.4.1.2 and 8.4.1.3, these shall be combined with operating procedures that ensure a dry-disconnect is achieved. The arrangement shall be subject to special consideration informed by a bunkering arrangement risk assessment conducted at the design stage and considering dynamic loads at the bunkering manifold connection to a recognized standard acceptable to the Administration, the safe operation of the ship and other hazards that may be r elevant to the ship during bunkering operation.

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ship and other hazards that may be r elevant to the ship during bunkering operation.
The fuel handling manual required by 18.2.3 shall include documentation that the bunkering arrangement risk assessment was conducted, and that special consideration was granted under this requirement.
8.4.3 An emergency release coupler (ERC) / Emergency Release System (ERS) or equivalent means shall be provided, unless installed on the bunkering supply side of the bunkering line, and the said means shall be in accordance with a standard equivalent to those acceptable to the Organization; it shal l enable a quick physical disconnection dry break-away of the bunker system in an emergency event."
9 Fuel supply to consumers
9.3 Regulations on redundancy of fuel supply
13 Paragraph 9.3.1 is replaced by the following:
"9.3.1 For ships constructed on or after 1 January 2026, for single fuel installations the fuel supply system shall be arranged with redundancy and segregation, so that a
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the fuel supply system shall be arranged with redundancy and segregation, so that a leakage in one system, or failure of one of the fuel supply essential auxiliaries, does not lead to an unacceptable loss of power. In the event of a leakage or failure, and in accordance with SOLAS regulation II -1/26.3, the Administration, having regard to overall safety considerations, may accept a partial reduction in propulsion capabili ty from normal operation."

9.4 Regulations on safety functions of gas supply system

14 Paragraph 9.4.7 is replaced by the following:

"9.4.7 For ships constructed on or after 1 January 2026, in cases where the master gas fuel valve is automatically shut down when the safety system as required in 15.2.2 is activated, the complete gas supply pipe between this master gas fuel valve and the

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is activated, the complete gas supply pipe between this master gas fuel valve and the double block and bleed valves and between the double block and bleed valves and the consumer shall be automatically vented."

15 Paragraph 9.4.8 is replaced by the following:

"9.4.8 For ships constructed on or after 1 January 2026, there shall be one manually operated shutdown valve in the gas supply line to each gas consumer upstream of the double block and bleed valves to assure safe isolation during maintenance on the gas consumer."

9.6 Regulations for fuel supply to consumers in gas-safe machinery spaces

16 Paragraph 9.6.1.1 is replaced by the following:

"9.6.1 Gas fuel piping in gas-safe machinery spaces shall be completely enclosed by a double pipe or duct fulfilling one of the following conditions:

.1 the gas fuel piping shall be a double wall piping system with the gas fuel contained in the inner pipe. The space between the concentric

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.1 the gas fuel piping shall be a double wall piping system with the gas fuel contained in the inner pipe. The space between the concentric pipes shall be pressurized with inert gas at a pressure greater than the gas fuel pressure. Suitable alarms shall be provided to indicate a loss of inert gas pressure between the pipes; or"

9.8 Regulations for the design of ventilated duct, outer pipe against inner pipe gas leakage

17 Paragraph 9.8.1 is replaced by the following:

"9.8.1 For ships constructed on or after 1 January 2026, the design pressure of the outer pipe or duct of fuel systems shall not be less than the maximum working pressure of the inner pipe. Alternatively, the design pressure of the outer pipe or duct may be calculated in accordance with 9.8.2."

18 The chapeau of paragraph 9.8.2 is replaced by the following:

"9.8.2 For ships constructed on or after 1 January 2026, alternatively to 9.8.1, the

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"9.8.2 For ships constructed on or after 1 January 2026, alternatively to 9.8.1, the design pressure of the outer pipe or duct shall be taken as the higher of the following:"

19 Paragraph 9.8.4 is replaced by the following:

"9.8.4 For ships constructed on or after 1 January 2026, the duct shall be pressure-tested to show that it can withstand the expected maximum pressure at fuel pipe rupture."

11 Fire safety

11.3 Regulations for fire protection

20 Paragraph 11.3.1 is replaced by the following:

"11.3.1 For ships constructed on or after 1 January 2026, fuel preparation rooms shall, for the purpose of the application of SOLAS regulation II-2/9, be regarded as a machinery space of category A."

11.6 Regulations for dry chemical powder fire-extinguishing system

21 Paragraph 11.6.2 is replaced by the following:

11.6 Regulations for dry chemical powder fire-extinguishing system
21 Paragraph 11.6.2 is replaced by the following:
"11.6.2 In addition to any other portable fire extinguishers that may be required elsewhere in IMO instruments, one portable dry powder extinguisher of at least 5 kg capacity shall be located near the bunkering station and in the fuel preparation room. For ships constructed before 1 January 2026, the portable dry powder extinguisher shall be provided in the fuel preparation room not later than the first survey on or after 1 January 2026."
12 Explosion prevention
12.5 Hazardous area zones
22 Paragraph 12.5.1 is replaced by the following:
"12.5.1 Hazardous area zone 0
For ships constructed on or after 1 January 2026, this zone includes, but is not limited to, the interiors of fuel tanks, any pipework for pressure relief or other venting systems for fuel tanks, pipes and equipment containing fuel, and interbarrier spaces as defined
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for fuel tanks, pipes and equipment containing fuel, and interbarrier spaces as defined
by paragraph 2.2.15.2."
12.5.2 Hazardous area zone 1
23 Paragraph 12.5.2.1 is replaced by the following:
".1 for ships constructed on or after 1 January 2026, tank connection spaces
and fuel storage hold spaces;"
15 Control, monitoring and safety systems
15.4 Regulations for bunkering and liquefied gas fuel tank monitoring
15.4.1 Level indicators for liquefied gas fuel tanks
24 Paragraph 15.4.1.3 is replaced by the following:
".3 For ships constructed on or after 1 January 2026, liquefied gas fuel tank
liquid level gauges may be of the following types:
.1 indirect devices which determine the amount of fuel by means such
as weighing or in-line flow metering;
.2 closed devices which do not penetrate the liquefied gas fuel tank,
such as devices using radioisotopes or ultrasonic devices; or
.3 closed devices which penetrate the liquefied gas fuel tank but which
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such as devices using radioisotopes or ultrasonic devices; or
.3 closed devices which penetrate the liquefied gas fuel tank but which form part of a closed system and keep the gas fuel from being released. Such devices shall be considered as tank connections. If the closed gauging device is not mounted directly onto th e tank, it shall be provided with a shut off valve located as close as possible to the tank."

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Part B-1
16 Manufacture, workmanship and testing
16.3 Welding of metallic materials and non -destructive testing for the fuel containment system
16.3.5 Production weld tests
25 Paragraph 16.3.5.1 is replaced by the following:
"16.3.5.1 For all fuel tanks and process pressure vessels except membrane tanks, production weld tests shall generally be performed for approximately
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tanks, production weld tests shall generally be performed for approximately each 50 m of butt -weld joints and shall be representative of each welding position. For secondary barriers, the same type production tests as required for primary barriers shall be performed, except that the number of tests may be reduced subject to agreement with the Administration. Tests, other than those specified in 16.3.5.2 to 16.3.5.5, may be required for fuel tanks or secondary barriers."

Part C-1

18 Operation

18.4 Regulations for bunkering operations

18.4.1 Responsibilities

26 Paragraph 18.4.1.1.1 is replaced by the following:

"18.4.1.1 Before any bunkering operation commences, the master of the receiving ship or their representative and the representative of the bunkering source (Persons In Charge, PIC) shall:

.1 agree in writing the transfer procedure, including cooling down and if necessary, gassing up; the maximum transfer rate at all stages ;

.1 agree in writing the transfer procedure, including cooling down and
if necessary, gassing up; the maximum transfer rate at all stages ;
minimum and maximum limiting transfer pressure and temperature;
bunkering line PRVs settings; and volume to be transferred;"
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